

TEST-AGENT

When Does Program Analysis Help LLM Test Generation?

An Adversarial, Cross-Model Study of Baseline vs. Agentic vs. Deep

COM4062 Graduation Project Report

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Supersedes the first-term pilot (Gemini, Jan 2026). This release reports the 100-task, 3-mode, multi-model study.

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1. Introduction — Problem Definition

Current success: Tools like Copilot generate code *fast*.

The cognitive gap — what LLMs still miss:

- Analyzing complex / adversarial failure scenarios
- Root-cause identification across files
- **Overconfidence:** believing *100% coverage = bug-free*

Core problem: A single LLM suffers from *hallucinations* and the *coverage illusion*. It lacks the self-reflection to verify its own tests.

A test that merely runs is not a test that reveals a bug.

1. Introduction — Aim of the Project

Goal: Build **TEST-AGENT** and answer a sharper question than "does an agent win?":

When — and for which models — does adding program analysis / an agentic loop actually improve a model's ability to write *bug-revealing* tests?

Investigation	Solution	Validation
Systematically probe LLM testing capability across the weak→strong spectrum	Compare three pipelines: Baseline → Agentic → Deep	Adversarial "trap" tasks + real bugs, judged by a deterministic oracle

Primary metric: **BRTR** (Bug-Revealing Test Rate). 100 tasks × 3 runs = 300 runs per cell.

2. Literature Review — The Evolution of Testing

Genetic Algorithms	Single LLMs	Agentic Testing
e.g. EvoSuite	e.g. Copilot	TEST-AGENT
High coverage, low readability	Simple errors, high hallucination risk	Cognitive loop, autonomous improvement

Research gap: The performance of LLMs in *adversarial (trap)* scenarios — and **whether the agentic loop helps or *hurts* depending on the model** — has not been systematically established.

Our contribution: a controlled, per-model measurement of that effect.

3. Method — Three Pipelines (the only variable)

The **same** TestWriter prompt, **same** retry budget, **same** deterministic validator are used everywhere. The only thing that changes is *how much analysis the model gets*:

Mode	Pipeline	Idea
Baseline	Task → TestWriter → Validate	Direct generation
Agentic	Task → Analyzer → TestWriter → Validate	Structured analysis injected
Deep	Task → tool-driven ReAct loop (read_file / run_tests) → Critic → Validate	Agent explores the code with tools

This fairness contract isolates a single variable: the **dose of analysis**.

3. Method — Agent Roles (Deep mode)

The Deep pipeline is a tool-augmented cognitive loop:

PLANNER	ANALYSIS	CRITIC	REFLECTION	EXECUTOR
Strategy & next step	Raw data → hypotheses	Refutes claims — the <i>overconfidence</i> blocker	Continue or complete?	Acts without interpreting

Core principles

- 1. **Agentic loop** — multi-stage cognitive cycle
- 2. **Blind-tool principle** — tools don't interpret; intelligence lives in the agent
- 3. **Deterministic orchestration** — reproducible runs

3. Method — The Bug-Revealing Oracle

A test counts as **bug-revealing** *if and only if*:

PASSES on the fixed (correct) code AND FAILS on the buggy code

- 100% deterministic — pytest return codes, **no LLM-as-judge**.
- Removes the "coverage illusion": a test that passes on the bug scores **zero**.
- Reported with **Wilson 95% confidence intervals** (300 runs per cell).

This single predicate is the source of truth for every number in this deck.

4. Experiments — Adversarial Benchmark

Trap scenarios where standard testing fails:

Coverage Illusion	Indirect Causation
Logic errors survive <i>100%</i> coverage	Error in File A, root cause in File B (config)

| **Silent Failures** | **State Dependency** |
| Swallowed exceptions / `except: pass` | Bugs that appear only in specific sequences |

Tasks drawn from custom adversarial traps + `BugsInPy`, `QuixBugs`, and `HumanEvalFix` real bugs.

4. Experiments — Setup & Metrics

Dataset	Models (full 3-mode trio)	Metrics
100 tasks (adversarial + BugsInPy / QuixBugs / HumanEvalFix)	6 models, weak → strong	BRTR (primary)
100 × 3 runs = 300 runs / cell	DeepSeek-V3.1 · V4-flash · gpt-oss-20b · llama-3.3-70b · llama-3.1-8b · phi-4	Wilson 95% CI
Per-task isolated workspaces	each run in all three modes	Token cost

Total: 5,400 controlled test runs (6 models × 3 modes × 300).

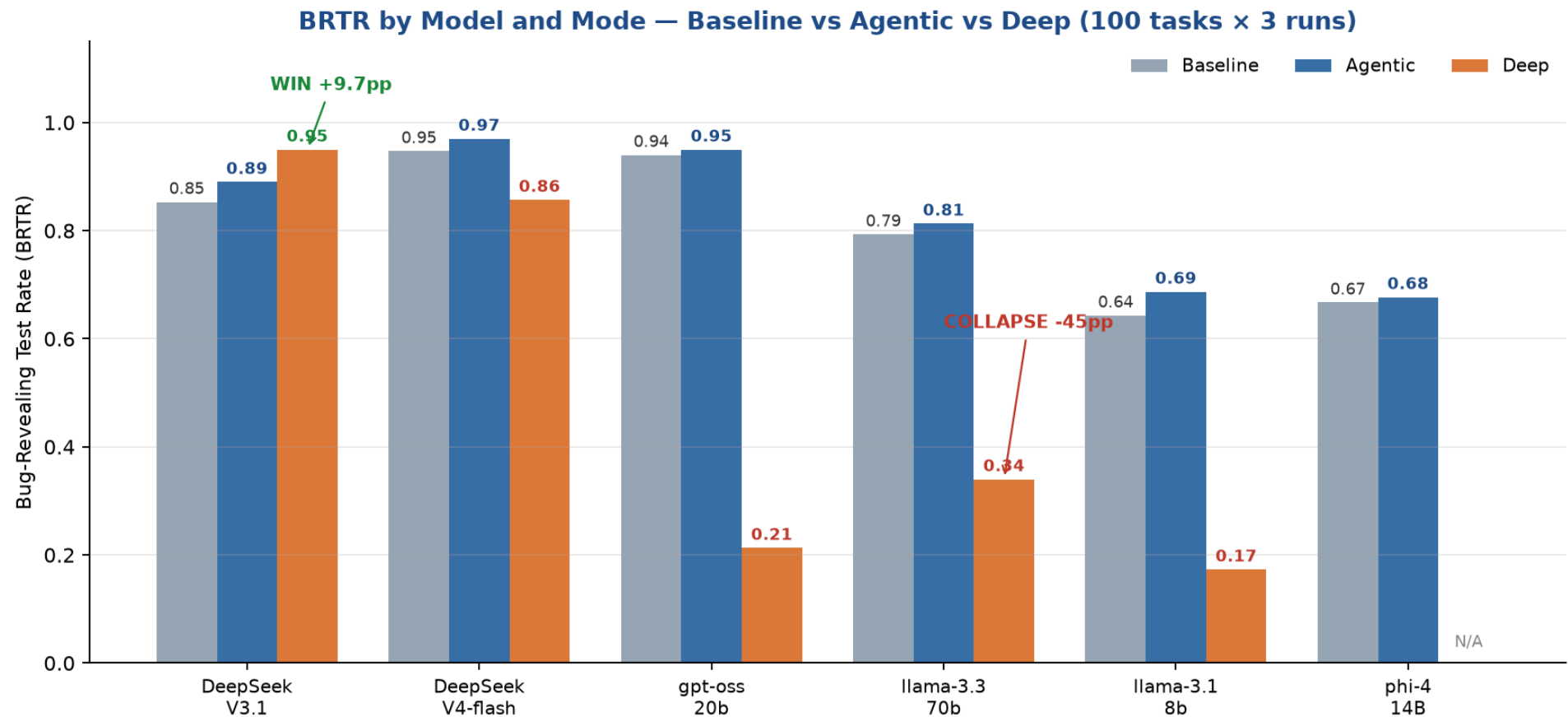
Models with a complete baseline + agentic + deep trio only — the rigorous comparison set.

5. Results — The Headline Table

Model	Baseline	Agentic	Deep	Δ (deep–base)
DeepSeek-V3.1	0.853	0.890	0.950	+9.7 pp ✓
DeepSeek-V4-flash	0.947	0.970	0.857*	≈ 0 (ceiling)
gpt-oss-20b	0.940	0.950	0.213	−72.7 pp ✨
llama-3.3-70b	0.793	0.813	0.340	−45.3 pp ✨
llama-3.1-8b	0.643	0.687	0.173	−47.0 pp ✨
phi-4 (14B)	0.667	0.677	N/A**	—

*V4-flash deep ≈ 0.973 once residual reasoning-harness stubs are excluded — neutral, not a collapse. **phi-4 deep unmeasurable: its provider endpoints offer no tool-calling (a provider limit, not a reasoning collapse).

5. Results — One Picture



Agentic $\approx$ baseline everywhere. **Deep** is the swing factor — it lifts V3.1, leaves V4-flash flat, *collapses* the weak / tool-clumsy models.

5. Results — Three Regimes of "Deep"

The analysis step is **not** uniformly good. It has three outcomes:

-  **Monotone benefit** — capable *and* not yet saturated:
DeepSeek-V3.1 0.853 → 0.890 → **0.950**. Deep's CI [91.9, 97.0] sits **entirely above** baseline [80.9, 88.9] → *statistically significant*.
-  **Ceiling / neutral** — already near the top:
DeepSeek-V4-flash (baseline 0.947) → deep $\approx$ baseline. No room to gain.
-  **Catastrophic collapse** — weak *or* tool-clumsy:
gpt-oss-20b (0.213), **llama-3.1-8b** (0.173), **llama-3.3-70b** (0.340).
The forced tool loop sabotages them; every failure is a *real* test failure.

6. The Killer Contrast — Capability ≠ Predictor

Two models with **almost the same baseline**, opposite deep outcome:

	Baseline	Deep	Outcome
DeepSeek-V3.1	0.853	0.950	climbs +9.7 pp
llama-3.3-70b	0.793	0.340	collapses −45.3 pp

Baseline capacity does **not** predict the deep result.
The 70B model is *large* yet collapses; the Llama family collapses at **both** 8B (−47 pp) and 70B (−45 pp).

The real predictor is agentic / tool-driving skill — DeepSeek drives the loop competently; Llama calls the tools but cannot follow the protocol.

6. Same Family, the "Headroom" Rule

Within one vendor family, the deep gain is governed by **how much room is left**:

DeepSeek	Baseline	Deep	Δ
V3.1 (has headroom)	0.853	0.950	+9.7 pp 
V4-flash (near ceiling)	0.947	0.857–0.973	≈ 0 (neutral)

Both drive the agentic loop well; the difference is purely **headroom**.

→ Analysis pays off only when the model is **both** unsaturated **and** agentially competent.

6. (Latest) A Fourth Architecture — Scout-Writer

If the collapse is the *dual burden* (drive tools **and** write the test in one loop), **separating** them should rescue the model:

- **Scout phase:** the model drives the tool loop to produce an *analysis only*.
- **Writer phase:** a fresh, tool-free context writes the test from that analysis.

gpt-oss-20b	Deep (one loop)	Scout-Writer
BRTR	0.213	0.897

Decoupling **repairs** the collapse (+68 pp) → the collapse was the *dual burden*, **not** the model's capability. (Neutral at the ceiling: Claude Haiku 0.987 / Sonnet 0.990.)

7. Conclusion

Achievements

- A controlled, 5,400-run, multi-model measurement of *when* analysis helps.
- **Positive result:** for a capable-but-unsaturated model (DeepSeek-V3.1), deep gives a **statistically significant** +9.7 pp.
- **Novel finding:** the deep outcome is predicted by **agentic skill, not raw capability** (V3.1 wins ↔ llama-3.3-70b collapses at the same baseline).
- A deterministic, no-judge oracle (BRTR + Wilson CI).

Limitations (honest)

- Deep is **double-edged**: it can *collapse* weak / tool-clumsy models.
- Cost: deep uses **~10–15× more tokens** and is far slower than baseline.

8. Future Work

Decouple	Route	Extend
Scout-Writer as a first-class mode — rescue the collapse band	Capability-aware routing — give deep only to models that gain from it	Beyond Python (Java, TypeScript); self-healing (test → fix)

Target the architecture at **low-baseline + tool-clumsy** models, where decoupling can convert the wasted deep budget into a real gain.

9. References

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Thank You

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Questions welcome.